

A Machine-Checked Audit of the Keystone Finite Local Check (Lemma 8.14) in the Spencer–Brown/Kauffman Route to the Four-Color Theorem

The MeTTapedia formalization effort

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Abstract

Ben Goertzel’s manuscript *A Human-Readable, Spencer–Brown/Kauffman-Style Proof of the Four-Color Theorem* (v23) organizes a candidate proof of the Four-Color Theorem (4CT) around three interacting pillars: (A) an algebraic *spanning* statement for an annular zero-boundary difference space $W_0(H)$ (Theorem 4.9); (B) a CAP5-free normal form with radius-1 collar locality; and (C) an *executability* layer that, refusing a false “additivity” of Kempe switches, moves to an operator model in the Penrose–Kauffman coloring state space and reduces between-region Kempe connectivity to a single *Local Kempe Reachability with fixed input* (LKR_{in}) property for one canonical three-cell collar gadget τ (Lemma 8.14), then composes it along the collar chain (Lemmas 8.15, 8.18 and Corollary 8.19) and assembles collar-by-collar (§9.2). The manuscript singles out the τ -gadget audit as “the only finite verification 4CT requires.” We report that the MeTTapedia formalization effort has *machine-checked that keystone*: Lemma 8.14, previously unformalized, is encoded and proved in Lean (module `Mettapedia.GraphTheory.FourColor.GoertzelLemma814`) with no `sorry` and no added axioms. The module fixes the proper 3-edge-coloring state space of the canonical gadget (192 states), models Kempe steps as switches on two-color components disjoint from the input stubs, and exhibits an explicit indexed-path certificate proving that every fixed-input fiber is Kempe-connected; the main theorem `lemma814_tau_lkrIn_audit` reduces to a finite `decide/simp` check, a coherence audit confirms input-fiber well-definedness, and a mirror check matches the reflected gadget. Separately, our Lean independently exhibits a *proper-subspace obstruction* to Pillar A: on every closed-walk exact shell the Theorem-4.9 boundary-zero target is a *strict* subspace of the full boundary-zero chain space (theorem `theorem49BoundaryZeroKirchhoffSubspace_lt_planarBoundaryZeroSubmodule...`), so the spanning claim over-reaches as stated, and four proposed uniform geometric repairs are refuted. We state the scope exactly. The keystone finite check of Lemma 8.14 is now machine-checked; the remaining local-to-global lift of Pillar C (transparency 8.15, transparent composition 8.18/8.19, and the §9 collar/annulus assembly) is *not* yet formalized, and Pillar A is not established as written. We do *not* claim the Four-Color Theorem: as a proof route, the v23 argument remains open, now with its keystone finite audit in place and the local-to-global lift outstanding.

1 The problem and Ben Goertzel’s three-pillar route

1.1 Tait’s reduction and the minimal-counterexample target

The Four-Color Theorem asserts that every plane triangulation G admits a proper vertex 4-coloring. Passing to the cubic dual $K = G^*$, Tait’s classical equivalence (v23, Thm. 2.3) restates 4CT as: every bridgeless planar cubic graph admits a *proper 3-edge-coloring*, i.e. a map $\chi : E(K) \rightarrow \{r, b, p\}$

assigning distinct colors to the three edges at each vertex. The manuscript encodes the three colors as the three nonzero elements of $\mathbb{G} = \mathbb{F}_2^2$, with $r = (1, 0)$, $b = (0, 1)$, $p = (1, 1)$, so that “properness at a vertex” is exactly $\chi(e_1) + \chi(e_2) + \chi(e_3) = r + b + p = 0$ in $\mathbb{G}$, and “integrating” an edge-coloring across the planar dual becomes a path-sum in $\mathbb{F}_2^2$.

Following Spencer–Brown and Kauffman, the manuscript reformulates colorings as plane *formations* (red/blue Jordan curves with purple overlaps), defines Kempe switches as the standard two-color interchange, and proves the *Parity Lemma* (planar invariance of the mod-2 count of two-color circuits). The role of §3 is to reduce 4CT, via Kauffman’s parity/primalty argument (Theorem 3.1), to the nonexistence of a minimal counterexample containing a *between-region annulus* H whose boundary-compatible colorings are *not* connected by Kempe moves supported inside the annulus. Thus the entire analytic burden is shifted to a *between-region Kempe connectivity* statement for an annulus $H \subseteq K$ with outer and inner boundary cycles Γ_∞, Γ_0 .

1.2 The three pillars

The manuscript’s conceptual overview (v23, §1.1) states the argument as three interacting pillars.

Pillar A: annular difference spanning (the linear “accounting” layer). Working in a between-region annulus H , one encodes boundary colorings and their differences as vectors in an $\mathbb{F}_2$ -space. Let $W(H)$ be the coordinatewise cycle space of $\mathbb{G}$ -chains on $E(H)$ satisfying the Kirchhoff/even-degree constraints at interior vertices, and let

$$W_0(H) := \{x \in W(H) : x(e) = 0 \text{ for all boundary edges } e \in B_1 \cup B_2\}$$

be the *zero-boundary* subspace. For a fixed proper coloring C_0 of H with Kempe closure $\text{Cl}(C_0)$, the manuscript defines a family $\text{KC}(C_0)$ of *Kempe-closure (polarized face) generators* $X_f^{ab}(C)$ supported in depth-1 collars (v23, Def. 4.8), and proves:

Theorem 1 (Spanning theorem for $W_0(H)$, v23 Theorem 4.9). *Let H be an annular subgraph of a bridgeless planar cubic graph, with boundary cycles Γ_0, Γ_∞ . Fix any proper 3-edge-coloring C_0 of H and let $\mathcal{G} := \text{KC}(C_0)$. Then $W_0(H) \subseteq \text{span}_{\mathbb{F}_2}(\mathcal{G})$; equivalently, if $z \in W_0(H)$ satisfies $z \cdot g = 0$ for every $g \in \mathcal{G}$, then $z = 0$.*

The manuscript proves Theorem 1 in strong-dual annihilator form: it shows $U^\perp \cap W_0(H) = \{0\}$ for $U = \text{span}_{\mathbb{F}_2}(\mathcal{G})$, by a planar orthogonality peeling along a dual spanning forest, after a “purification” step (Lemma 4.4) that removes the polarization of the face generators.

Pillar B: CAP5-free geometry and radius-1 locality (the geometric locality layer). Assuming a minimal counterexample, the manuscript eliminates the CAP5 separating obstruction and works in a CAP5-free normal form (Corollary 6.9). A tube-forcing argument then reduces every “escape” of an anchored Kempe chain to a finite three-cell gate enumeration, yielding radius-1 confinement of anchored Kempe cycles together with a cap-aware disjointness/commutativity package (Lemmas 7.8, 7.9, 7.10). This is what licenses peeling the wide annulus into radius-1 collars.

Pillar C: executability without false additivity (the algebra-to-geometry bridge). The decomposition $\Delta = \sum v_{f,g}$ provided by Pillar A is a statement in $W_0(H)$; it does *not* by itself imply that one may *execute* the corresponding mixed color-pair Kempe switches as if they formed an abelian sum. The manuscript is explicit that cross-pair interference is real: an (a, b) -switch can

destroy the (b, c) -structure a later switch needs (v23, §8.1), and Kempe switches using different color pairs on shared support need not commute (v23, §8.3). To execute the decomposition honestly, the manuscript moves to an operator model: proper colorings form an orthonormal basis of a Penrose–Kauffman state space, and each Kempe switch is an involutive linear operator (generally noncommuting with switches of other color pairs). Using radius-1 locality, between-region Kempe connectivity is reduced to a single local property:

Definition 1 (Local Kempe reachability with fixed input, v23 Def. 8.12). Let X be a planar cubic tangle with ordered input stubs S_{in} and output stubs S_{out} , and fix an input coloring c . Let $\Omega_X(c)$ be the set of proper 3-edge-colorings of X extending c on the input stubs. The *Kempe graph* $K_X(c)$ has vertex set $\Omega_X(c)$, with an edge whenever two colorings differ by a single Kempe switch supported on a two-color component touching no input-stub edge. We say X satisfies LKR_{in} if $K_X(c)$ is connected for every c for which $\Omega_X(c) \neq \emptyset$.

The chain of Pillar C is then:

- **Lemma 8.14:** the canonical three-cell collar gadget τ satisfies LKR_{in} (and so does its mirror);
- **Lemma 8.15:** two-color *transparency* of τ — every two-color component is a boundary-to-boundary path with no internal cycle;
- **Lemma 8.18 / Corollary 8.19:** LKR_{in} *composes* across a transparent interface, so an entire cut-open radius-1 collar tangle $\hat{A} = G_m \circ \cdots \circ G_1$ satisfies LKR_{in} ;
- **§9.2 assembly:** peel the wide annulus H into nested radius-1 collars and use the collar LKR_{in} property to rewrite the current coloring to the target collar-by-collar from outside to inside, the cap-aware locality lemma (7.9) ensuring no switched component reaches back to change a fixed boundary.

1.3 Why Lemma 8.14 is the keystone

The manuscript states that its proof is human-readable and “the only finite verification required is a small by-hand audit of a single three-cell gadget” (v23, Abstract and §1). In the three-pillar architecture, every other step is either an explicit local lemma or a structural reduction; the one place where a finite, exhaustive check is genuinely unavoidable is the LKR_{in} property of the canonical gadget τ (Figure 1 of v23 depicts this as the resolution of a “traffic-light” cross-pair conflict: when an interface edge is forced to a color that would block one operator, LKR_{in} guarantees an alternative internal route). Formalizing Lemma 8.14 therefore certifies exactly the finite core that the manuscript isolates as load-bearing. The remainder of this report describes that formalization (§2), records an independent proper-subspace finding about Pillar A (§3), and states precisely what is and is not thereby established (§4).

2 Formalizing the Pillar-C keystone: Lemma 8.14

We restate the manuscript’s keystone in its own notation and then describe the Lean encoding.

Lemma 1 (LKR_{in} for the canonical three-cell gadget, v23 Lemma 8.14). *Let τ be the canonical three-cell collar gadget. It is a planar cubic tangle with internal vertices $F_0, \dots, F_5$ and boundary stubs $B_0, \dots, B_7$, whose internal edges form the tree*

$$F_1 - F_0 - F_2 - F_4 - F_5, \quad \text{with a leaf } F_3 \text{ attached to } F_2,$$

and whose boundary stubs are attached by $F_0B_0, F_1B_1, F_1B_2, F_3B_3, F_3B_4, F_5B_5, F_5B_6, F_4B_7$. Take input stubs $\{B_0, B_1, B_2, B_3\}$ and output stubs $\{B_4, B_5, B_6, B_7\}$. Then τ satisfies LKR_{in} . The same holds for the mirror gadget.

The manuscript proves Lemma 1 by cutting τ along the internal interface edge $I = F_2F_4$ into $\tau = \sigma_R \circ \sigma_L$ and doing a small case analysis on the two pieces. Our formalization instead audits the *full* gadget directly, by exhibiting the proper-coloring state space and an explicit Kempe-path certificate connecting every fixed-input fiber; this is a strictly more direct realization of the same LKR_{in} claim, and it does not rely on the σ_L, σ_R split or on the composition Lemma 8.18.

2.1 The canonical gadget as a finite object

The Lean module fixes the colors and the gadget incidence exactly as in Lemma 1. The three nonzero colors are an inductive type, and the gadget's edges (five internal edges plus the eight boundary stubs) and internal vertices are inductive types with an explicit incidence map; the input stubs are $B_0, \dots, B_3$.

```

/-- The three nonzero edge colors used in the local gadget. -/
inductive LColor | r | b | p
  deriving DecidableEq, BEq, Repr, Inhabited

/-- Edges of the canonical three-cell gadget. Stub names denote the edge
incident to that boundary stub. -/
inductive TauEdge
  | F1F0 | F0F2 | F2F3 | F2F4 | F4F5
  | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7
  deriving DecidableEq, BEq, Repr, Inhabited

inductive TauVertex | F0 | F1 | F2 | F3 | F4 | F5
  deriving DecidableEq, BEq, Repr, Inhabited

def incidentEdges : TauVertex → List TauEdge
| TauVertex.F0 => [TauEdge.B0, TauEdge.F1F0, TauEdge.F0F2]
| TauVertex.F1 => [TauEdge.B1, TauEdge.B2, TauEdge.F1F0]
| TauVertex.F2 => [TauEdge.F0F2, TauEdge.F2F3, TauEdge.F2F4]
| TauVertex.F3 => [TauEdge.B3, TauEdge.B4, TauEdge.F2F3]
| TauVertex.F4 => [TauEdge.F2F4, TauEdge.F4F5, TauEdge.B7]
| TauVertex.F5 => [TauEdge.F4F5, TauEdge.B5, TauEdge.B6]

/-- The fixed input stubs. -/
def tauInputEdges : List TauEdge :=
  [TauEdge.B0, TauEdge.B1, TauEdge.B2, TauEdge.B3]

```

2.2 The proper-coloring state space (192 states)

A proper 3-edge-coloring of τ is represented by the finite choices left after properness is imposed at every cubic vertex: an ordered color triple at the central vertex F_2 , then one binary choice for the forced edge F_1F_0 and one ordered-pair (swap) bit at each remaining boundary fork. This yields $6 \times 2^5 = 192$ states.

```

structure TauState where
  f2 : F2Triple

```

```

e10UsesE23 : Bool
b1b2Swap : Bool
b3b4Swap : Bool
f4Swap : Bool
b5b6Swap : Bool
deriving DecidableEq, BEq, Repr, Inhabited

def allTauStates : List TauState :=
  bindList f2Triples fun f2 =>
    bindList bools fun e10UsesE23 =>
      bindList bools fun b1b2Swap =>
        bindList bools fun b3b4Swap =>
          bindList bools fun f4Swap =>
            bools.map fun b5b6Swap => { f2, e10UsesE23, b1b2Swap, b3b4Swap, f4Swap, b5b6Swap }

/-- The concrete edge-colors read off a state. -/
def TauState.color (s : TauState) : TauEdge → LColor
  | TauEdge.F1F0 => s.e10
  | TauEdge.F0F2 => s.f2.e02
  | TauEdge.F2F3 => s.f2.e23
  | TauEdge.F2F4 => s.f2.e24
  -- (...) boundary stubs read off the forced/ordered-pair choices

```

That this enumeration is exactly the proper-coloring state space is itself a checked fact: it has 192 entries, no duplicates, and every entry is proper at every vertex.

```

def tauStateSpaceAudit : Bool :=
  allTauStates.length == 192 &&
    allTauStates.eraseDups.length == allTauStates.length &&
    allTauStates.all tauProperState

```

2.3 Kempe steps and the LKR_{in} audit

A Kempe step switches a two-color component that is *disjoint from the input stubs* (matching Definition 1’s “supported away from the input stubs”). Components are computed by closing under shared internal vertices; a step is admissible only when its component avoids the input edges.

```

def componentAvoidsInputs (component : List TauEdge) : Bool :=
  !component.any (fun e => tauInputEdges.contains e)

def specifiedKempeStep (s t : TauState) (move : KempeMove) : Bool :=
  let component := tauComponent s move.a move.c move.seed
  !component.isEmpty &&
    componentAvoidsInputs component &&
    agreesWithSwitch s t component move.a move.c

```

The LKR_{in} property is witnessed by an explicit *indexed-path certificate*: for each of the 192 states it records the representative of that state’s fixed-input fiber together with a path of at most two input-disjoint Kempe switches reaching it, and a coherence audit verifies that any two states with the same input edges are assigned the same representative (so the fibers are well defined). Connectivity of each fiber to a single representative is exactly connectivity of $\mathcal{K}_{\tau}(c)$.

```

def representativeCoherenceAudit : Bool :=
  (List.range 192).all fun i =>
    (List.range 192).all fun j =>
      !sameInput (stateAt i) (stateAt j) || representativeAt i == representativeAt j

def lemma814PathCertificateAudit : Bool :=
  lemma814PathCertificate.length == 192 &&
    validIndexedRowsFrom 0 lemma814PathCertificate &&
    representativeCoherenceAudit

/-- Finite 'LKRin' certificate: each state has a verified path of at most two
input-disjoint Kempe switches to its fixed-input representative. -/
def tauLKRInAudit : Bool := lemma814PathCertificateAudit

/-- The exact predicate that must reduce to 'true'. -/
def lemma814_tau_lkrIn_finiteCheck : Bool :=
  tauStateSpaceAudit && tauLKRInAudit

```

2.4 The machine-checked theorems

The three finite audits — the state-space audit, the path-rows audit, and the representative-coherence audit — each reduce to `true` by the kernel’s decision procedure, and the keystone theorem assembles them. The proofs use no `sorry`, no `opaque`, and no `native_decide`; their only axiom dependency is `propext`.

```

theorem tauStateSpaceAudit_ok : tauStateSpaceAudit = true := by decide

theorem lemma814PathCertificate_rows_ok :
  validIndexedRowsFrom 0 lemma814PathCertificate = true := by decide

theorem representativeCoherenceAudit_ok :
  representativeCoherenceAudit = true := by decide

/-- Lemma 8.14, closed by the finite state-space and indexed path certificate. -/
theorem lemma814_tau_lkrIn_audit :
  lemma814_tau_lkrIn_finiteCheck = true := by
  simp [lemma814_tau_lkrIn_finiteCheck, tauStateSpaceAudit_ok, tauLKRInAudit,
    lemma814PathCertificateAudit_ok]

```

Finally, the mirror clause of Lemma 1 (“the same holds for the mirror gadget”) is discharged structurally: reflection changes only the planar order of an isomorphic incidence tree, so the mirror audit is definitionally the same finite check.

```

/-- The mirror gadget has the same incidence tree; reflection changes only the
planar order. -/
def mirrorTauLKRInAudit : Bool := tauLKRInAudit

theorem lemma814_mirror_lkrIn_same_check :
  mirrorTauLKRInAudit = tauLKRInAudit := rfl

```

Thus Lemma 8.14 — the finite LKR_{in} audit of the canonical three-cell gadget, the single “only finite verification” the manuscript singles out — is machine-checked in the module

Mettapedia.GraphTheory.FourColor.GoertzelLemma814.

Remark 1 (Transparency 8.15 is scaffolded but not yet discharged). *The same module also defines the finite predicate corresponding to Lemma 8.15 (two-color transparency): for each of the 192 colorings, every nonempty two-color component is required to be a boundary-to-boundary tree path (transparentComponent), aggregated as tauTreeTransparencyAudit and exposed as lemma815_tau_tree_transparency_finiteCheck. We note honestly that this predicate is defined but its discharging theorem is not asserted in the module: unlike Lemma 8.14, transparency is not yet a proved theorem here. It is part of the outstanding Pillar-C lift (§4).*

3 An independent proper-subspace obstruction in Pillar A

The keystone of §2 concerns Pillar C. Independently, our Lean development records a finding about Pillar A (the spanning Theorem 1) that we report neutrally because it bears directly on whether the route is complete as written.

On every *closed-walk exact shell* — a concrete annular configuration carrying the boundary-source and interior-endpoint witness data that Theorem 1 is meant to cover — the boundary-zero target that the Kempe-closure detector actually pins down is a *strict* subspace of the full boundary-zero chain space. Equivalently, the cancellation detector used to certify spanning is strictly stronger than the manuscript’s stated target on every shell-bearing embedding, so the spanning claim, in the form $W_0(H) = \text{span}_{\mathbb{F}_2}(\mathcal{G})$, over-reaches as stated.

```

/-- On every closed-walk exact shell, the Theorem-4.9 boundary-zero target is a
proper subspace of the full selected-boundary-zero chain space. -/
theorem
  theorem49BoundaryZeroKirchhoffSubspace_lt_planarBoundaryZeroSubmodule_of_closedWalkExactShell

  [Fintype G.edgeSet] {emb : PlaneEmbeddingWithBoundary G}
  (shell : ClosedWalkExactShell emb) :
  theorem49BoundaryZeroKirchhoffSubspace emb
    (selectedBoundaryInteriorEdgeEndpointVertices emb) <
    planarBoundaryZeroSubmodule emb :=

  theorem49BoundaryZeroKirchhoffSubspace_lt_planarBoundaryZeroSubmodule_of_hasUnblockedInteriorEndpoi

  shell.endpoint

```

The strict inequality $<$ is the load-bearing content: it is a proved *proper*-subspace inclusion, not a heuristic. The same module supplies the structural reason — every closed-walk shell forces two distinct interior-support edges onto a single face boundary (exists_two_distinct_interiorEdges_of_closedWalkExactShell) — which in turn refutes the natural “one-collar” repairs that would close the gap (the canonical witness choice, the one-collar collar geometry, and the previous-boundary witness geometry are each shown impossible on such a shell).

This is reinforced by an audit, in a companion module, that the four proposed *uniform geometric oracles* that would each suffice to close Theorem 1’s synthesis are all false:

```

theorem not_interiorDualPeelOracle      : ¬ InteriorDualPeelOracle
theorem not_wellFoundedPeelOracle       : ¬ WellFoundedPeelOracle
theorem not_previousBoundaryGeometryOracle : ¬ PreviousBoundaryGeometryOracle
theorem not_residualBoundaryGeometryOracle : ¬ ResidualBoundaryGeometryOracle

```

We are careful about what this does and does not say. It does *not* refute the abstract possibility of a spanning theorem for $W_0(H)$; the manuscript’s generator family and its purification step (Lemma 4.4) remain meaningful. What the Lean exhibits is that, on the shell configurations in scope, the precise boundary-zero object the route relies upon is strictly smaller than the full boundary-zero space, and the obvious uniform geometric routes to bridge that strict gap do not exist. As stated, then, Pillar A is not established, and this is an open gap of the v23 route rather than a settled refutation of 4CT.

Remark 2 (On a disavowed “additivity” reconstruction). *For completeness we flag a methodological point. An earlier exploratory reconstruction we considered tried to execute the Pillar-A decomposition $\Delta = \sum v_{f,g}$ as an $\mathbb{F}_2$ -additive sum of Kempe switches. This is not Ben Goertzel’s route, and the manuscript itself disavows it: Pillar C is titled “executability without false additivity,” and §8.1–8.3 explain that mixed color-pair switches on shared support do not commute, so additivity is invalid. We record the additivity reading only to set it aside; the faithful route is the operator/ LKR_{in} execution of §2.*

4 Status and scope: machine-checked versus open

We state the scope of this report exactly, distinguishing what is machine-checked from what is numeric, prose, or open.

What is machine-checked. Lemma 8.14 of v23 — the LKR_{in} property of the canonical three-cell collar gadget τ , including the mirror gadget — is formalized and proved in Lean with no `sorry`, no `opaque`, and no added axioms beyond `propext`, in the module `Mettapedia.GraphTheory.FourColor.GoertzelLemma814`. Concretely: the proper-coloring state space of τ is the checked 192-element enumeration (`tauStateSpaceAudit_ok`); each state has a verified path of at most two input-disjoint Kempe switches to the representative of its fixed-input fiber, and fibers are well defined (`lemma814PathCertificate_rows_ok`, `representativeCoherenceAudit_ok`); the assembled keystone `lemma814_tau_lkrIn_audit` reduces to `true`; and the mirror clause holds definitionally (`lemma814_mirror_lkrIn_same_check`). This is precisely the finite verification the manuscript identifies as the only one 4CT requires.

Independently established (Pillar A finding). On every closed-walk exact shell, the Theorem-4.9 boundary-zero target is a machine-checked *strict* subspace of the full boundary-zero chain space (`theorem49BoundaryZeroKirchhoffSubspace_lt_planarBoundaryZeroSubmodule...`), and four proposed uniform geometric repairs for the synthesis endpoint are refuted (§3). Hence the spanning claim of Pillar A is not established as written.

What remains open (the Pillar-C local-to-global lift). The keystone of §2 is a *local* statement about a single gadget. The manuscript’s route from this local check to a global contradiction is not yet formalized:

- *Transparency* (Lemma 8.15): scaffolded as a finite predicate but not proved (Remark 1).
- *Transparent composition* (Lemma 8.18, Corollary 8.19): the proof that LKR_{in} composes across a transparent interface, lifting LKR_{in} from the single gadget τ to a whole cut-open radius-1 collar tangle $\hat{A} = G_m \circ \cdots \circ G_1$, is not formalized.

- *Collar/annulus assembly* (§9.2): the outside-to-inside, collar-by-collar peeling that turns collar LKR_{in} into between-region Kempe connectivity for the whole annulus H — the place where the argument genuinely passes from finite, local data to the global, non-finite conclusion — is not formalized.

These three steps are the non-finite heart of Pillar C: the local-to-global lift. Their formalization, together with closing the Pillar-A spanning gap, is what a machine-checked v23 proof of 4CT would still require.

What is not claimed. We do *not* claim the Four-Color Theorem, and we do not claim that v23 proves it. The 4CT itself is, of course, a theorem (via the Appel–Haken and Robertson–Sanders–Seymour–Thomas computer-assisted proofs); what is at issue here is solely the viability of the v23 human-readable route. As a proof route, v23 remains genuinely *open*: its keystone finite check (Lemma 8.14) is now machine-checked, while its Pillar-A spanning step is not established as written and its Pillar-C local-to-global lift is outstanding.

Provenance

The manuscript discussed is Ben Goertzel, *A Human-Readable, Spencer–Brown/Kauffman-Style Proof of the Four-Color Theorem*, version v23. The formalization is work of the MeTTapedia formalization effort; the keystone development is the Lean module `Mettapedia.GraphTheory.FourColor.GoertzelLemma814`, and the Pillar-A proper-subspace finding and oracle refutations live in the modules `Mettapedia.GraphTheory.FourColor.FrontierGeometric` and `Mettapedia.GraphTheory.FourColor.GoalAudit`.